# PRODUCT REQUIREMENT DOCUMENT (PRD)

## UMHackathon 2026

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**Project Name:** Cook.GPT

## 1. Project Overview

- **Problem Statement:** The "Decision Paralysis" of a cluttered kitchen is a hidden financial drain. Currently, Malaysian households lose significant money weekly due to food spoilage, largely because visualizing a meal from random, mismatched ingredients requires specialized culinary "Inference" skills.
- **Target Domain: Domestic Resource Optimization & Household FinTech.** We are operating at the intersection of Food-Tech and Decision Intelligence, specifically targeting busy urban professionals and students (like those at UM) who need to maximize their grocery ROI under time pressure.
- **Proposed Solution Summary: Cook.GPT** removes the friction of "What's for dinner?" by transforming a simple photo into a structured financial recovery plan. By leveraging the **Z.AI ILMU-GLM-5.1** vision model, we shift the user's role from **Manual Laborer** (typing lists) to **Strategic Curator** (choosing between AI-synthesized, zero-waste meal options).

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## 2. Background & Business Objective

- **Background of the Problem:**
  For decades, managing household food inventory has relied on a **manual, high-friction linear pathway**:
  1. **The Audit:** A user manually checks the fridge and tries to remember what they bought.
  2. **The Search:** The user searches for recipes based on those ingredients (often finding recipes that require *even more* ingredients).
  3. **The Loss:** If the "mental load" of planning is too high, the ingredients are ignored until they expire.

  This process requires high "culinary literacy" and significant time—skills and resources that many modern urban Malaysians lack. This specialized knowledge barrier prevents people from being economically efficient in their own kitchens.

- **Importance of Solving This Issue:**
  - **Democratizing Expertise:** We break the barrier of culinary skills; the AI acts as a "Senior Chef" who knows exactly how to substitute missing ingredients (e.g., using soy sauce + sugar for oyster sauce).
  - **Economic Recovery:** It transforms potential waste back into wealth. If every Malaysian household saves just RM 10/week using Cook.GPT, the collective national economic impact is in the millions.
- **Strategic Fit / Impact:**

  **Cook.GPT** aligns perfectly with the **Z.AI "Decision Intelligence" Ecosystem**.

  - **Integration:** We utilize the **ILMU-GLM-5.1** model not as a side-chat, but as the primary engine for image-to-economic inference.
  - **Localization:** Our solution focuses on the **ASEAN Market**, specifically training our prompts to recognize local Malaysian ingredients.
  - **AI-Native Lifecycle:** This project demonstrates how an AI-Native approach can replace traditional database-heavy applications with a lightweight, "Inference-First" architecture, reducing the need for massive manual data entry.

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## 3. Product Purpose

### 3.1. Main Goal of the System:

The primary objective of **Cook.GPT** is to serve as an AI-driven Economic Decision Engine for the kitchen. It aims to eliminate the "cognitive load" and "financial friction" associated with food waste by:

- **Converting Visual Chaos to Structured Value**: Using computer vision to instantly audit fridge assets.
- **Synthesizing Inference-Based Solutions**: Generating dynamic, culturally relevant recipes that prioritize "at-risk" ingredients.
- **Empowering Resource Stewardship**: Shifting the user's role from a passive consumer to an active curator of their household's domestic economy.

### 3.2. Intended Users (Target Audience):

- **Urban Professionals:** Busy individuals in cities like KL who have "high-value time" and need to minimize grocery-run frequency while maintaining a healthy diet.

- **University Students (e.g., UM Students):** Budget-conscious users who need to maximize the shelf-life of limited ingredients to save money.

- **"Last-Minute" Meal Planners:** Users who frequently face "Decision Paralysis" at the end of the day and risk letting food spoil due to lack of inspiration.

- **Sustainability Advocates:** Eco-conscious consumers focused on reducing their carbon footprint by achieving a "Zero-Waste" household.

- **Culinary Beginners:** Individuals who lack the "Chef's intuition" to perform smart ingredient substitutions and need an AI to perform that inference for them.

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## 4. System Functionalities

### 4.1. Description:

The system operates as a **Vision-to-Economic-Inference engine** designed to optimize household food assets. Unlike traditional search-based recipe apps, Cook.GPT utilizes a **Multi-Modal Generative Model** to capture visual relationships between ingredients, calculate perishability risks, and synthesize dynamic culinary solutions that prioritize financial recovery.

### 4.2. Key Functionalities:

- **Visual Pantry Auditing:** Translates unstructured images of fridge/pantry shelves into a structured inventory list with near-zero user manual entry.
- **Economic Risk Scoring:** Automatically assigns a Ringgit (RM) value to identified items and categorizes them by urgency (URGENT, SOON, OK) based on inferred shelf-life.
- **Dynamic Recipe Synthesis:** "Inferences" 2 to 3 culturally authentic Malaysian recipes on the fly, utilizing **Smart Substitution** logic to ensure a 100% "Zero-Waste" outcome based *only* on available ingredients.
- **Impact Dashboard:** A persistent financial tracker that visualizes "Total RM Saved" and "XP Level," converting mundane household chores into a gamified economic win.

### 4.3. AI Model & Prompt Design

#### 4.3.1. Model Selection: Z.AI (ILMU-GLM-5.1)

We selected the **ILMU-GLM-5.1** model because of its superior **spatial reasoning** and **multi-modal vision capabilities**. Specifically:

- **High-Resolution Object Detection:** It accurately identifies localized Malaysian produce (e.g., *Cili Padi* vs. regular *Chili*) which generic models often misidentify.
- **Structured Output Consistency:** It excels at following strict **JSON schemas**, which is vital for our "Inference" paradigm where the frontend must parse precise recipe steps and RM values.

#### 4.3.2. **Prompting Strategy: Multi-Step Agentic Prompting**

We utilize a Multi-Step Agentic approach rather than zero-shot prompting:

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1. Step 1 (Vision): "Identify all food items in this image and estimate their freshness."
2. Step 2 (Economic Calculation): "Cross-reference the identified list with the provided Market Price Array."
3. Step 3 (Synthesis): "Create a recipe that rescues the highest-value expiring items using only the detected inventory + specified pantry staples." This ensures the AI doesn't "hallucinate" ingredients the user doesn't actually own.

4.3.3. **Context & Input Handling**:

- **Maximum Input:** The system accepts images up to **10MB**.
- **Pre-Processing:** Images are downscaled to **512x512 pixels** at the edge before transmission.
- **Chunking/Rejection:** If the visual context window is exceeded, the system performs "Region-of-Interest" cropping. Non-food images are rejected at the header level by a lightweight classification filter to save API costs.

4.3.4. **Fallback & Failure Behavior:**

- **Schema Validation:** If the GLM returns malformed JSON, a **middleware retry** is triggered (up to 2 times).
- **Graceful Error State:** If the model returns an unusable response after retries, the system falls back to a **"Safe-Mode Gallery"** of pre-cached recipes based on the user's most common past ingredients.
- **Human-in-the-loop:** Users can "Flag" an incorrect detection, which sends the data to our **User Correction Ledger** for future prompt tuning.

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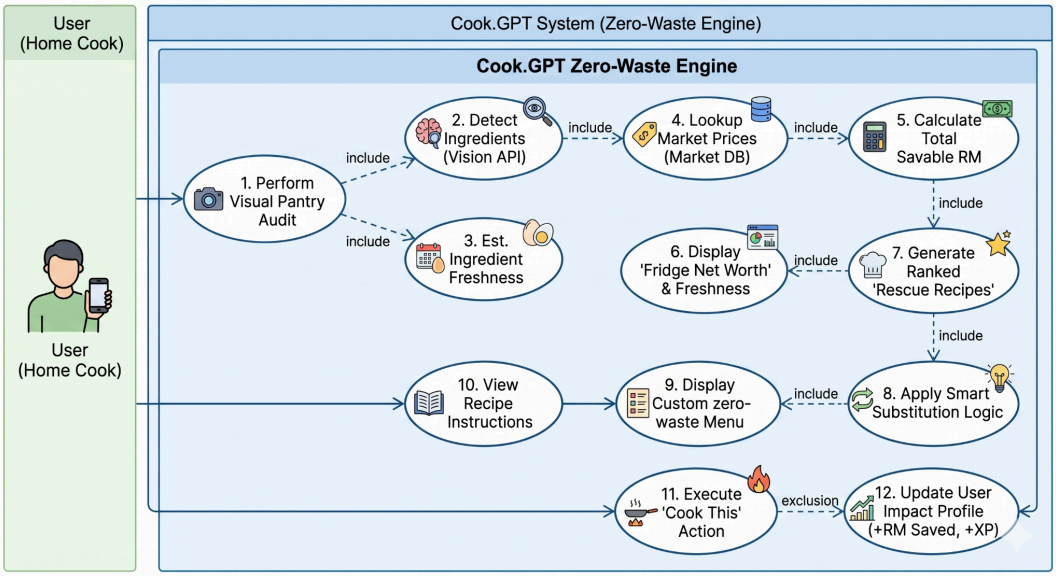
## 5. User Stories & Use Cases

- **User Stories:**
  - As a university student on a tight budget, I want to take a photo of my leftover ingredients so that the app can tell me exactly what to cook and how much money I just saved from not buying takeout

- **Use Cases (Main Interactions):**
  - **Visual Ingestion Flow:** User snaps a photo, Cook.GPT parses the image, categorizes freshness, and displays the "Total Savable RM."
  - **Dynamic Rescue Flow**: User reviews the "Ranked Meal Options," reads the Smart Substitution rationale, and selects a recipe to view step-by-step interactive instructions.
  - **Gamification Loop**: User clicks "Cook This," triggering an in-app toast notification. The RM value and XP are permanently added to their Impact Profile.

- **Use Case Diagram:**

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**6. Features Included (Scope Definition)**

- **Generative Visual Auditor:** A mobile-first interface that enables users to capture or upload photos of their kitchen inventory. The system performs multi-object detection to translate these pixels into a structured digital inventory list.

- **Real-Time Economic Valuator:** An automated logic layer that assigns estimated Ringgit (RM) values to detected ingredients based on a localized Malaysian market price database, providing an instant "Fridge Net Worth."

- **Inference-Based Recipe Synthesizer:** A dynamic engine that "rescues" at-risk ingredients by generating 2-3 culturally authentic recipes (e.g., *Nasi Goreng Kampung*, *Ayam Masak Merah*). It utilizes **Smart Substitution logic** to fill gaps using only the user's identified pantry staples.

- **Interactive Impact Dashboard:** A persistent tracking module that aggregates the total financial value of food "rescued" across sessions, awarding **XP** and **Chef Levels** to gamify the user's journey toward a zero-waste lifestyle.

**7. Features Not Included (Scope Control)**

- **Real-Time Grocery Integration:** While the "Grand Vision" includes one-click purchasing from partners like Jaya Grocer or Lotus's, the MVP will not feature live checkout or delivery logistics.

- **IoT Smart Fridge Synchronization:** The system will not integrate with "Smart Fridge" internal cameras or external sensors. All inventory data is derived from user-initiated photo uploads.

- **Social & Community Sharing:** Features such as "Friend Feed," "Community Recipe Sharing," or "Social Cooking Challenges" are deferred to post-hackathon versions to focus on the private household economic utility.

- **Personalized Health/Nutritional Tracking:** The engine prioritizes **economic rescue** and **culinary synthesis**. Extensive caloric tracking, macro-nutrient breakdowns, or integration with fitness wearables (Apple Health/Google Fit) are currently out of scope.

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**8. Assumptions & Constraints**

- **LLM Cost Constraint:**
  - Vision inference is token-heavy. To keep costs manageable, the application utilizes highly specific bounding-box prompts and restricts the output generation to 2-3 highly targeted recipes rather than an exhaustive list.

- **Technical Constraints:**
  - The app relies on React (Vite) and requires a stable internet connection for the Z.AI API calls.
  - Exporting grocery lists to third-party apps is currently unsupported.

- **Performance Constraints:**
  - Processing high-res images through Vision LLMs causes inherent latency. We designed a cinematic 10-second loading UI to mask this processing time as a "diagnostics phase," turning a technical limitation into an engaging UX feature.

- **User Input:**

  - The accuracy of the RM valuation and recipes is heavily dependent on the lighting and clarity of the photo provided by the user.

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**9. Risks & Questions Throughout Development**

- **Safety & Health**
  - How do we prevent the AI from recommending the consumption of completely rotten food?
  - Mitigation: Prompt engineering strictly directs the model to classify visibly spoiled food as hazardous, focusing only on "wilting" or "soon-to-expire" items.
- **Culinary Accuracy**
  - Will the AI generate strange flavor combinations?
  - Mitigation: The prompt restricts generation to established Southeast Asian flavor profiles and traditional cooking methods.
- **User Retention**
  - Will users scan their fridge every day?
  - Mitigation: Implemented the XP and Leveling system to create a dopamine-driven feedback loop, gamifying food waste reduction.

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